

1st (TS) Philippine Mathematical Olympiad

Area Stage, 31 October 2025

PART I. The answer to each item is an integer from 1 to 999. No solution is needed. Write your answers in the answer sheet. Each correct answer is worth three points.

1. Kaiser's favorite two-digit number is such that subtracting 2 or 4 from it results in a number that shares exactly one digit with the original, but subtracting 3 results in a number that shares no digits with the original. Find Kaiser's favorite number.
2. A set of primes is called *complete* if the digits of its elements are $1, 2, \dots, 9$ in some order. For example, the set $\{7, 83, 421, 659\}$ is complete. Compute the smallest possible sum of elements of a complete set.
3. In parallelogram $ABCD$, the angle bisectors of $\angle A$ and $\angle B$ intersect segment CD and divide the interior of the parallelogram into four regions: two triangles and two quadrilaterals. Given that the triangles have areas 27 and 75, find the area of $ABCD$.
4. Compute the number of 6-digit positive integers $\overline{ABCDEF}$ satisfying:
 - $\overline{AB}$ is divisible by 4.
 - $\overline{ABCD}$ is divisible by 16.
 - $\overline{ABCDEF}$ is divisible by 64.
5. The graphs of $y = x^3 - 5x^2 + 7x + 4$ and $y = 2x^2 - 6$ intersect at A , B , and C . If the sum of the coordinates of the centroid of triangle ABC can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.
6. In regular polygon $A_1A_2 \cdots A_n$ for a positive integer $n \geq 25$, define X as the reflection of A_1 across A_3 . Given that X , A_5 , and A_{25} are collinear, find n .
7. The positive divisors of 1001 are randomly written on four two-sided cards, one on each face. If the probability that on every card, one number divides the other can be expressed as $\frac{m}{n}$, find $m + n$.
8. Let ABC be a triangle with centroid G . Suppose the reflection of A across B lies on the circumcircle of BGC such that $GB = 3$ and $GC = 9$. If the area of ABC can be expressed as $m\sqrt{n}$ for square-free n , find $m + n$.
9. Each tile of a 3×3 chessboard is colored either white or black at random. If the probability that a knight can move from the bottom left tile to the top right tile without changing colors or a knight can move from the bottom right tile to the top left tile without changing colors, but not both can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.
10. How many integers $1 < n \leq 1000$ satisfy the property that $p^2 + 2$ divides n , where p is the smallest prime factor of n ?
11. Let ABC be a triangle with $AB = 13$, $BC = 14$, $CA = 15$, and circumcenter O . Suppose AO intersects the circumcircle of BOC at $X \neq O$. If the length of AX can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.
12. Let n be the smallest positive integer such that

$$3 \left\lfloor \frac{n}{3} \right\rfloor - 4 \left\lfloor \frac{n}{4} \right\rfloor = 7 \left\lfloor \frac{n}{7} \right\rfloor - 8 \left\lfloor \frac{n}{8} \right\rfloor = 11 \left\lfloor \frac{n}{11} \right\rfloor - 12 \left\lfloor \frac{n}{12} \right\rfloor = 1.$$

Find the remainder when n is divided by 1000.

13. Let f be a function defined by $f(1) = 1$, $f(30) = 0$, and for all $x \neq 1, 30$,

$$f(x) = \sum_{\text{prime } p|x} f\left(\frac{x}{p}\right).$$

Find the remainder when $f(1620000)$ is divided by 1000.

14. Acute triangle ABC has circumcenter O and circumradius 81. Let line ℓ be the tangent to the circumcircle of ABC at A such that the distances from B and C to ℓ are 64 and 144 respectively. If the area of ABC can be expressed as $m\sqrt{n}$ for square-free n , find the remainder when $m + n$ is divided by 1000.
15. For each nonnegative integer k , $P_k(x)$ is a real polynomial of degree k with leading coefficient $\frac{k+1}{k!}$. For each k , we have that

$$P_k(k) = P_{k+1}(k) = P_{k+2}(k) = \cdots$$

Given that $P_{100}(100)$ is an integer, compute the number of positive integer divisors of $P_{100}(100)$.

PART II. Write your solutions to each problem in the solution sheets. Each complete and correct solution is worth ten points.

- Chick and Rick play a game on n points spaced equally on a circle. Chick goes first, and on each turn the player draws a chord on the circle between some pair of the points. The first player to draw a chord intersecting an existing chord strictly in the interior of the circle loses. Find, with proof, all positive integers n such that Chick wins when the game is played optimally.
- Find, with proof, a non-recursive formula for the number of sequences $s_1, s_2, \dots, s_N$ such that $s_1 = 1$, $s_2 = 2$, $s_3 = 3$, and for all integers $3 < k \leq N$,

$$s_k \in \{s_{k-1}, s_{k-2}, s_{k-3}\}.$$

- A sequence $\{a_n\}_{n \geq 1}$ of positive integers is defined recursively by $a_1 = k$ and

$$\begin{cases} a_{n+1} = \frac{1}{2} \cdot a_n & \text{if } a_n \equiv 0 \pmod{2} \\ a_{n+1} = a_n + 77 & \text{if } a_n \equiv 1 \pmod{2} \end{cases}$$

for all n . Determine, with proof, the sum of all values of k for which $a_6 = a_1$.

Part I.

- | | | |
|-----------------------------------|-----------------------------------|---------------------------------------|
| 1. 82 | 6. 48 | 11. $271 \left(\frac{260}{11}\right)$ |
| 2. 207 | 7. $25 \left(\frac{4}{21}\right)$ | 12. 449 (1449) |
| 3. 240 | 8. $29 (27\sqrt{2})$ | 13. 890 (64890) |
| 4. 308 | 9. $37 \left(\frac{5}{32}\right)$ | 14. 186 $(5184\sqrt{2})$ |
| 5. $62 \left(\frac{59}{3}\right)$ | 10. 181 | 15. 404 |

1. **(2025 FLSAM Intro Meeting, Jacob Khohayting)** Kaiser's favorite two-digit number is such that subtracting 2 or 4 from it results in a number that shares exactly one digit with the original, but subtracting 3 results in a number that shares no digits with the original. Find Kaiser's favorite number.

Answer. 82

Solution. Notice, the units digit must be at most 2, since if it is greater than 2 then subtracting 3 yields the same tens digit. Suppose the units digit is 2 first, say $\overline{n2}$. Then, subtracting 2 yields that the tens digit is the same. Subtracting 3 and 4 yields $\overline{(n-1)9}$ and $\overline{(n-1)8}$. Considering subtracting 4, then $n-1=2$ or $n=8$. If $n-1=2$, then the tens digit upon subtracting 3 is the same as the units digit of the original number. Thus, $n=8$. So, the number is 82 which works as subtracting 2, 3, and 4 yield 80, 79, and 78. This solution ensures uniqueness. Thus, the answer is 82.

2. **(2025 FLSAM Intro Meeting, Jingyi Li)** A set of primes is called *complete* if the digits of its elements are 1, 2, ..., 9 in some order. For example, the set $\{7, 83, 421, 659\}$ is complete. Compute the smallest possible sum of elements of a complete set.

Answer. 207

Solution. Notice, 4, 6, and 8 are all even numbers. If they were in the units digit of some number, then that value would be divisible by 2. Hence, these digits cannot be in the units digit.

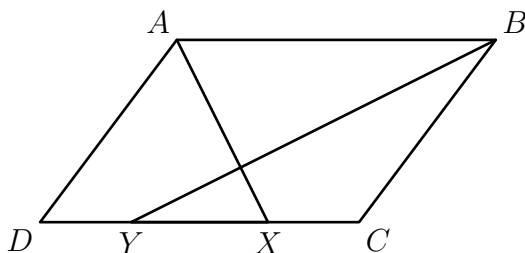
So, if we can somehow place the remaining numbers in the units digit positions, then we would minimize the sum. In fact, we can with the construction $\{2, 3, 5, 41, 67, 89\}$. Thus, the minimum possible sum is:

$$2 + 3 + 5 + 41 + 67 + 89 = \boxed{207}.$$

3. **(2025 FLSAM PUMaC Tryout, Jacob Khohayting)** In parallelogram $ABCD$, the angle bisectors of $\angle A$ and $\angle B$ intersect segment CD and divide the interior of the parallelogram into four regions: two triangles and two quadrilaterals. Given that the triangles have areas 27 and 75, find the area of $ABCD$.

Answer. 240

Solution.



Let P be the intersection of the angle bisectors of $\angle A$ and $\angle B$. Let X and Y be the intersection of angle bisectors of $\angle A$ and $\angle B$ with CD , respectively.

Easily, we find:

$$[ABCD] = 2 \cdot [ABC] = 2 \cdot \left(\frac{AB \cdot d(C, AB)}{2} \right) = AB \cdot (d(P, CD) + d(P, AB))$$

But, $\triangle APB \sim \triangle YPX$ with ratio $\frac{5}{3}$. Thus, $d(P, AB) = \frac{5}{3} \cdot d(P, CD)$.

$$[ABCD] = AB \cdot \left(d(P, AB) \cdot \frac{8}{5} \right) = \left(\frac{1}{2} \cdot AB \cdot d(P, AB) \right) \cdot \frac{16}{5} = [APB] \cdot \frac{16}{5} = 75 \cdot \frac{16}{5} = \boxed{240}.$$

4. **(2025 FLSAM ARML Tryout, Jacob Khohayting)** Compute the number of 6-digit positive integers $\overline{ABCDEF}$ satisfying:

- $\overline{AB}$ is divisible by 4.
- $\overline{ABCD}$ is divisible by 16.
- $\overline{ABCDEF}$ is divisible by 64.

Answer. $\boxed{308}$

Solution. Let $X = \overline{AB}$, $Y = \overline{CD}$, and $Z = \overline{EF}$. We can express the six-digit positive integer as $10^4 \cdot X + 10^2 \cdot Y + Z$, where A , B , and C are two-digit positive integers. For the integer to be divisible by 64, the following conditions must be met:

- X must be divisible by 4.
- $10^2 \cdot X + Y$ must be divisible by 16.
- $10^4 \cdot X + 10^2 \cdot Y + Z$ must be divisible by 64.

Since $10^2 \cdot X$ is already divisible by 16, Y must be divisible by 16. Similarly, since $10^4 \cdot X + 10^2 \cdot Y$ is divisible by 64, Z must be divisible by 64. The possible values for X are $\{12, 16, \dots, 96\}$, giving us 22 options. The possible values for Y are $\{00, 16, 32, \dots, 96\}$, providing 7 options. For Z , the possible values are 00 and 64, resulting in 2 options. Multiply yields $22 \cdot 7 \cdot 2 = \boxed{308}$.

5. **(2025 FLSAM PUMaC Tryout, Jacob Khohayting)** The graphs of $y = x^3 - 5x^2 + 7x + 4$ and $y = 2x^2 - 6$ intersect at A , B , and C . If the sum of the coordinates of the centroid of triangle ABC can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.

Answer. $\boxed{62}$

Solution. Let (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) be the points of intersection. Then, x_1 , x_2 , and x_3 must satisfy

$$x^3 - 5x^2 + 7x + 4 = 2x^2 - 6 \iff x^3 - 7x^2 + 7x + 10 = 0.$$

Thus, $x_1 + x_2 + x_3 = 7$ and $x_1x_2 + x_2x_3 + x_3x_1 = 7$. Therefore,

$$\begin{aligned} \sum_{\text{cyc}} y_i &= \sum_{\text{cyc}} (2x_i^2 - 6) = 2 \cdot \sum_{\text{cyc}} x_i^2 - 18 \\ &= 2 \cdot \left(\left(\sum_{\text{cyc}} x_i \right)^2 - \left(\sum_{\text{cyc}} x_i x_j \right) \right) - 18 = 2 \cdot (7^2 - 2 \cdot 7) - 18 = 52. \end{aligned}$$

Hence, the sum of the coordinates of the centroid is equivalent to

$$\frac{1}{3} \cdot \left(\sum_{\text{cyc}} x_i + \sum_{\text{cyc}} y_i \right) = \frac{59}{3}.$$

Thus, $59 + 3 = \boxed{62}$.

6. **(2024 FLSAM FLOMO, Jacob Khohayting)** In regular polygon $A_1A_2\cdots A_n$ for a positive integer $n \geq 25$, define X as the reflection of A_1 across A_3 . Given that X , A_5 , and A_{25} are collinear, find n .

Answer. 48

Solution. Notice, $\triangle XA_3A_5 \sim \triangle XA_{25}A_1$. Thus,

$$\begin{aligned}\angle XA_3A_5 &= \angle XA_{25}A_1 = \angle A_5A_{25}A_1 = 180 \cdot \frac{4}{n}, \text{ and} \\ \angle XA_5A_3 &= \angle XA_1A_{25} = \angle A_5A_1A_{25} = 180 \cdot \frac{22}{n}.\end{aligned}$$

Hence, $\angle A_3XA_5 = (1 - \frac{26}{n}) \cdot 180$. In addition, $A_3A_5 = A_1A_3 = XA_3$, so $\angle XA_5A_3 = \angle A_5XA_3$. Therefore, $1 - \frac{26}{n} = \frac{22}{n}$ and $n = \boxed{48}$.

7. **(2025 FLSAM PUMaC Tryout, Jacob Khohayting)** The positive divisors of 1001 are randomly written on four two-sided cards, one on each face. If the probability that on every card, one number divides the other can be expressed as $\frac{m}{n}$, find $m + n$.

Answer. 25

Solution. Firstly, there are $\binom{8}{4} \cdot 4! \cdot \frac{1}{2^4} = 105$ total ways to label the four cards. Notice, 7, 11, and 13 are all on different cards. Now, we will perform casework:

- If 1 is paired with 1001, then $7 \cdot 11$, $11 \cdot 13$, and $13 \cdot 7$ are all paired with one of 7, 11, and 13. If $7 \cdot 11$ is paired with 7, then $7 \cdot 13$ must be paired with 13 and $11 \cdot 13$ must be paired with 11. A similar result occurs if $7 \cdot 13$ is paired with 7. There are a total of two ways for this case.
- If 1 isn't paired with 1001, then since no two of $7 \cdot 11$, $11 \cdot 13$, and $13 \cdot 7$ can be paired with one another, select exactly one of them to be paired with either 1 or 1001. Say, we select $7 \cdot 11$. We will multiply by 3 at the end. There are 2 ways to select the other side to $7 \cdot 11$.
Now, if $11 \cdot 13$ pairs with 11, there are 2 ways for $13 \cdot 7$. But, if $11 \cdot 13$ pairs with 13, then $7 \cdot 13$ must pair with 7. Thus, yielding one way.

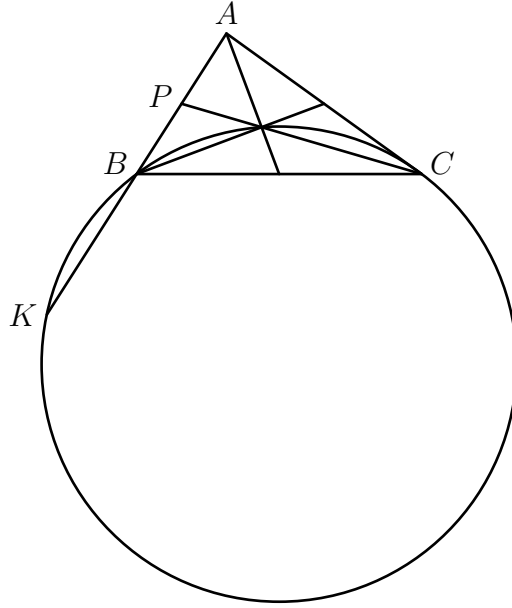
Hence, there are $3 \cdot 2 \cdot (2 + 1) = 18$ ways for this case.

In total, there are 20 ways. Hence, the probability that on every card, one number divides the other is $\frac{20}{105} = \frac{4}{21}$ yielding $4 + 21 = \boxed{25}$.

8. **(2025 FLSAM PUMaC Tryout, Jacob Khohayting)** Let ABC be a triangle with centroid G . Suppose the reflection of A across B lies on the circumcircle of BGC such that $GB = 3$ and $GC = 9$. If the area of ABC can be expressed as $m\sqrt{n}$ for square-free n , find $m + n$.

Answer. 29

Solution.



Let $P = CG \cap AB$; that is, P is the midpoint of AB . By Power of Point, we find:

$$PB \cdot PK = PG \cdot PC \implies PB \cdot (3 \cdot PB) = PG \cdot (3 \cdot PG) \implies PB = PG.$$

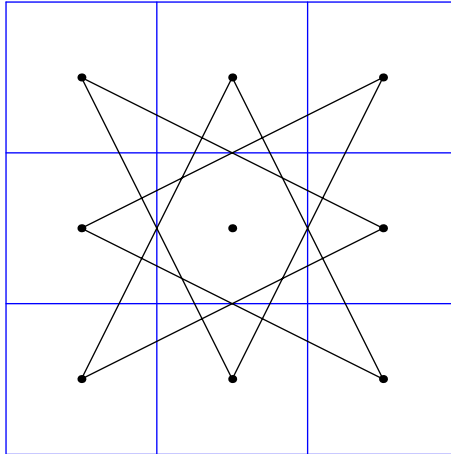
Thus, P is the center of (ABG) . Thus, $\angle ABG = 90^\circ$ and $AB = 9$. By Pythagorean Theorem, then we find $AG = 6\sqrt{2}$. Now, we may now compute the area of $\triangle ABC$:

$$[ABC] = 3 \cdot [ABG] = 3 \cdot \frac{AG \cdot BG}{2} = 3 \cdot \frac{6\sqrt{2} \cdot 3}{2} = 27\sqrt{2} \implies \boxed{29.}$$

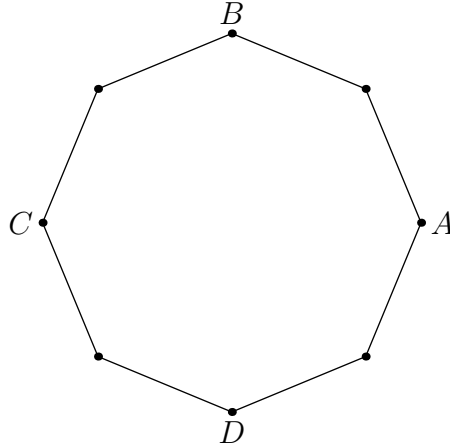
9. **(2025 FLSAM Intro Meeting, Ben Jiang)** Each tile of a 3×3 chessboard is colored either white or black at random. If the probability that a knight can move from the bottom left tile to the top right tile without changing colors or a knight can move from the bottom right tile to the top left tile without changing colors, but not both can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.

Answer. $\boxed{37}$

Solution. We will compute the probability that the knight can get from the bottom left to the top right but not from the bottom right to the top left; the desired probability is twice this. Consider the graph on the tiles generated by knight moves:



Note that the knight can never get to the center tile. “Unwrapping” the graph, it becomes a cycle:



Here, A , B , C , and D represent the bottom left, bottom right, top right, and top left tiles respectively. Suppose the knight can get from A to C via the upper arc through B . This requires A , C , and the 3 tiles on the arc to all be the same color, which occurs with probability $\frac{1}{24}$. For the knight to not be able to get from B to D , we can check that 5 of the 8 ways to color the 3 tiles on the lower arc work. Adding the case where the knight can get from A to C via the lower arc, noting that these cases are disjoint, we get

$$2 \cdot \frac{1}{24} \cdot \frac{5}{8}$$

Then the desired probability is

$$2 \cdot 2 \cdot \frac{1}{24} \cdot \frac{5}{8} = \frac{5}{32}.$$

Thus, $5 + 32 = \boxed{37}$.

10. **(2023 FLSAM PUMaC Tryouts, Aaron Hu)** How many integers $1 < n \leq 1000$ satisfy the property that $p^2 + 2$ divides n , where p is the smallest prime factor of n ?

Answer. $\boxed{181}$

Solution. If $2 \mid n$, then $6 \mid n$. Then, there are $\lfloor \frac{1000}{6} \rfloor = 166$ total ways.

If $3 \mid n$, then $11 \mid n$ and $n \nmid 2$, so $n \equiv 1 \pmod{2}$. Thus, we require for n to satisfy:

$$\begin{cases} n \equiv 1 \pmod{2} \\ n \equiv 0 \pmod{3} \\ n \equiv 0 \pmod{11} \end{cases}$$

Equivalently, by Chinese Remainder Theorem, $n \equiv 33 \pmod{66}$ yielding $\lfloor \frac{1000+33}{66} \rfloor = 15$ ways.

We will show no other solutions exists. Suppose $p > 3$ is the smallest prime. Note the unique non-zero square in $\mathbb{Z}_3$ is 1. Hence, if $p > 3$, then $3 \mid (p^2 + 2)$. Then, the smallest prime is at most 3, a contradiction. Hence, no solutions exist if $p > 3$.

Therefore, there are exactly $166 + 15 = \boxed{181}$ total possible values of n .

11. **(2025 FLSAM Intro Meeting)** Let ABC be a triangle with $AB = 13$, $BC = 14$, $CA = 15$, and circumcenter O . Suppose AO intersects the circumcircle of BOC at $X \neq O$. If the length of AX can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.

Answer. $\boxed{271}$

Solution. Since AH and AO are isogonal, then $\angle BAH = \angle XAC$. Also, BO and BH are isogonal. In conjunction with cyclic quadrilateral $BOCX$, then:

$$\angle AXC = \angle OXC = \angle OBC = \angle HBA.$$

Therefore, $\triangle ABH \sim \triangle AXC$. Thus, we obtain $\frac{AB}{AH} = \frac{AX}{AC}$. Then we can compute

$$AX = \frac{AB \cdot AC}{AH} = \frac{bc}{2R \cdot \cos(A)} = \frac{13 \cdot 15}{2 \cdot \frac{65}{8} \cdot \frac{33}{65}} = \frac{260}{11}.$$

This yields $260 + 11 = \boxed{271}$.

12. **(2025 FLSAM PUMaC Tryouts, Jacob Khohayting)** Let n be the smallest positive integer such that

$$3 \left\lfloor \frac{n}{3} \right\rfloor - 4 \left\lfloor \frac{n}{4} \right\rfloor = 7 \left\lfloor \frac{n}{7} \right\rfloor - 8 \left\lfloor \frac{n}{8} \right\rfloor = 11 \left\lfloor \frac{n}{11} \right\rfloor - 12 \left\lfloor \frac{n}{12} \right\rfloor = 1.$$

Find the remainder when n is divided by 1000.

Answer. $\boxed{449}$

Solution. Consider the solutions for a general a :

$$a \left\lfloor \frac{m}{a} \right\rfloor - (a+1) \left\lfloor \frac{m}{a+1} \right\rfloor = 1.$$

Suppose $m = ak + r$ for $0 \leq r < a$, then: $a \left\lfloor \frac{m}{a} \right\rfloor = ak$. But, $ak \equiv 1 \pmod{a+1}$. But, $a \equiv -1 \pmod{a+1}$, so $k \equiv -1 \pmod{a+1}$. Thus, $k = a \pmod{a+1}$. Then, $a^2 \leq m \pmod{a^2+a} < a^2 + a$. Hence, we find the set of solutions to each of the equalities are:

- $3 \left\lfloor \frac{m}{3} \right\rfloor - 4 \left\lfloor \frac{m}{4} \right\rfloor \implies \{9, 10, 11\} \pmod{12}$.
- $7 \left\lfloor \frac{m}{7} \right\rfloor - 8 \left\lfloor \frac{m}{8} \right\rfloor \implies \{49, 50, 51, \dots, 55\} \pmod{56}$.
- $11 \left\lfloor \frac{m}{11} \right\rfloor - 12 \left\lfloor \frac{m}{12} \right\rfloor \implies \{121, 122, 123, \dots, 131\} \pmod{132}$.

Notice,

$$\{9, 10, 11\} \pmod{12} \cap \{121, 122, 123, \dots, 131\} \pmod{132} = \{129, 130, 131\} \pmod{132}.$$

Let $m = 132a - k$ where $k \in \{1, 2, 3\}$. Then:

$$\begin{aligned} 132a - k \equiv \{49, 50, 51, \dots, 55\} \pmod{56} &\iff 20a - k \equiv \{49, 50, 51, \dots, 55\} \pmod{56} \\ &\iff 20a \equiv \{49 + k, 50 + k, 51 + k, \dots, 55 + k\} \pmod{56} \\ &\iff 20a \equiv \{50, 51, 52, \dots, 58\} \pmod{56}. \end{aligned}$$

So,

$$20a = 56b + \{50, 51, 52, \dots, 58\} \iff \{18, 17, 16, \dots, 10\} - 4b \equiv 0 \pmod{20}.$$

We want to minimize a . In turn, this implies we want to minimize b . To minimize, we require $b = 3$. Hence: $20a = 56b + \{50, 51, 52, \dots, 58\} = \{218, 219, 220, \dots, 226\}$. From here, it isn't hard to see the minimal a which satisfies this is $a = 11$. Thus,

$$132 \cdot 11 - k \equiv \{49, 50, 51, \dots, 55\} \pmod{56}.$$

But, $132 \cdot 11 \equiv 52 \pmod{56}$. Hence, to minimize m , we want to maximize k such that $132 \cdot 11 - k$ still satisfies the above equality. As $k \in \{1, 2, 3\}$, we take $k = 3$ which yields the desired minimal m of $132 \cdot 11 - 3 = 1449$. Hence, the answer is $\boxed{449}$.

Alternate Solution. (Better) Let $r_m(n)$ denote the remainder when n is divided by m . We have

$$3 \left\lfloor \frac{n}{3} \right\rfloor - 4 \left\lfloor \frac{n}{4} \right\rfloor = 1 \implies \left(n - 4 \left\lfloor \frac{n}{4} \right\rfloor \right) - \left(n - 3 \left\lfloor \frac{n}{3} \right\rfloor \right) = 1 \implies r_4(n) - r_3(n) = 1$$

Similarly, $r_8(n) - r_7(n) = r_{12}(n) - r_{11}(n) = 1$.

Let $k = r_3(n)$. Then $r_4(n) = k+1$. By Chinese Remainder Theorem, $r_{12}(n) = k+9$, and so $r_{11}(n) = k+8$. Now, we have that $r_8(n)$ is one of $k+1, k+5$.

If $r_8(n) = k+5$, then $r_7(n) = k+4$ and

$$n \equiv k-3 \pmod{1848}$$

where $1848 = \text{lcm}(3, 4, 7, 8, 11, 12)$. This yields a smallest possible n of $1848-3$, which is too large.

Otherwise, $r_8(n) = k+1$, $r_7(n) = k$, and by Chinese Remainder Theorem

$$n \equiv k+1449 \pmod{1848}$$

This yields a smallest possible n of $\boxed{1449}$, achieved at $k=0$.

13. **(2024 FLSAM PUMaC Tryouts, Darsh Patel & Xuzhou Ren)** Let f be a function defined by $f(1) = 1$, $f(30) = 0$, and for all $x \neq 1, 30$,

$$f(x) = \sum_{\text{prime } p|x} f\left(\frac{x}{p}\right).$$

Find the remainder when $f(1620000)$ is divided by 1000.

Answer.

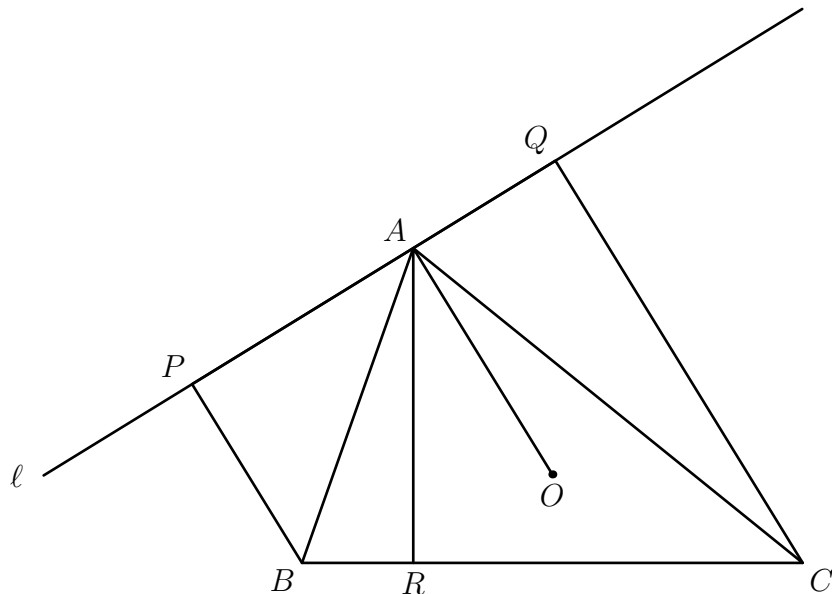
Solution. Notice that there is a bijection from the value of $f(1620000)$ to the number of paths from $(0, 0, 0)$ to $(5, 4, 4)$ not passing through $(1, 1, 1)$. This is simply

$$\binom{13}{5, 4, 4} - \binom{3}{1, 1, 1} \binom{10}{4, 3, 3} = \frac{13!}{5!4!4!} - \frac{3!}{1!1!1!} \cdot \frac{10!}{4!3!3!} = 64 \boxed{890}.$$

14. **(2025 FLSAM PUMaC Tryouts, Jacob Khohayting)** Acute triangle ABC has circumcenter O and circumradius 81. Let line ℓ be the tangent to the circumcircle of ABC at A such that the distances from B and C to ℓ are 64 and 144 respectively. If the area of ABC can be expressed as $m\sqrt{n}$ for square-free n , find the remainder when $m+n$ is divided by 1000.

Answer. $\boxed{186}$

Solution.



Let P and Q be the feet of the perpendiculars from B and C to ℓ , respectively. In addition, let R be the foot of the perpendicular from A to BC .

Notice, $\triangle ABP \sim \triangle CAR$ and $\triangle BAR \sim \triangle QCA$. Therefore:

$$\begin{aligned} \frac{AB}{PB} = \frac{AC}{AR} &\implies \frac{AB}{64} = \frac{AC}{AR} \implies 64 = \frac{AB \cdot AR}{AC} \\ \frac{AC}{QC} = \frac{AB}{AR} &\implies \frac{AC}{144} = \frac{AB}{AR} \implies 144 = \frac{AC \cdot AR}{AB}. \end{aligned}$$

Hence, we may find:

$$\frac{144}{64} = \frac{(AC \cdot AR)/AB}{(AB \cdot AR)/AC} = \left(\frac{AC}{AB}\right)^2 \implies \frac{AC}{AB} = \frac{3}{2}.$$

Thus, $AR = 96$. Let $AB = 2a$ and $AC = 3a$. Also, let $BC = s$. Notice, $OA = d(O, \ell) = 81$. Therefore, using the circumradius formula, we find:

$$[ABC] = \frac{AB \cdot BC \cdot CA}{4 \cdot OA} = \frac{2a \cdot BC \cdot 3a}{4 \cdot 81}.$$

But, $[ABC] = \frac{1}{2} \cdot BC \cdot AR$. Thus:

$$\frac{1}{2} \cdot BC \cdot 96 = \frac{2a \cdot BC \cdot 3a}{324} \implies \frac{96 \cdot BC}{2} = \frac{6a^2 \cdot BC}{324} \implies a = 36\sqrt{2}.$$

Thus, we find:

$$BC = \sqrt{(AB)^2 - (AR)^2} + \sqrt{(AC)^2 - (AR)^2} = \sqrt{(72\sqrt{2})^2 - 96^2} + \sqrt{(108\sqrt{2})^2 - 96^2} = 108\sqrt{2}.$$

But, as $AR = 96$, then $[ABC] = 5184\sqrt{2}$ yielding $\boxed{186}$.

15. **(2025 FLSAM PUMaC Tryouts, Darsh Patel)** For each nonnegative integer k , $P_k(x)$ is a real polynomial of degree k with leading coefficient $\frac{k+1}{k!}$. For each k , we have that

$$P_k(k) = P_{k+1}(k) = P_{k+2}(k) = \dots$$

Given that $P_{100}(100)$ is an integer, compute the number of positive integer divisors of $P_{100}(100)$.

Answer. $\boxed{404}$

Solution. Let $c_k = k + 1$, so that $P_k(x)$ has leading coefficient $\frac{c_k}{k!}$. Let $a_k = P_k(k)$. Then we are given that for each k ,

$$(a_0, a_1, \dots, a_k) = (P_k(0), P_k(1), \dots, P_k(k))$$

Note that $P_k(x) - P_{k-1}(x)$ has roots at $0, 1, \dots, k-1$. This implies that

$$P_k(x) = P_{k-1}(x) + \frac{c_k}{k!}x(x-1)\cdots(x-k+1) = P_{k-1}(x) + c_k\binom{x}{k}.$$

Thus,

$$P_k(x) = c_k\binom{x}{k} + c_{k-1}\binom{x}{k-1} + \dots + c_0\binom{x}{0}.$$

We now compute

$$\begin{aligned} P_{100}(100) &= 101\binom{100}{100} + 100\binom{100}{99} + \dots + 1\binom{100}{0} \\ &= \left(\binom{100}{100} + \binom{100}{99} + \dots + \binom{100}{0}\right) + 100\left(\binom{99}{99} + \binom{99}{98} + \dots + \binom{99}{0}\right) \\ &= 2^{100} + 100(2^{99}) \\ &= 2^{100} \cdot 3 \cdot 17. \end{aligned}$$

Then the desired value is $\boxed{404}$.

Part II.

1. **(2025 FLSAM Intro Meeting, Aaryan Vaishya)** Chick and Rick play a game on n points spaced equally on a circle. Chick goes first, and on each turn the player draws a chord on the circle between some pair of the points. The first player to draw a chord intersecting an existing chord strictly in the interior of the circle loses. Find, with proof, all positive integers n such that Chick wins when the game is played optimally.

Solution. The game is forced to draw exactly the n sides of the polygon and $n - 3$ diagonals in a triangulation, so there are always $2n - 3$ moves and Chick always wins. Hence, n is all positive integers.

2. **(2025 FLSAM PUMaC Tryouts, Jacob Khohayting)** Find, with proof, a non-recursive formula for the number of sequences $s_1, s_2, \dots, s_N$ such that $s_1 = 1$, $s_2 = 2$, $s_3 = 3$, and for all integers $3 < k \leq N$,

$$s_k \in \{s_{k-1}, s_{k-2}, s_{k-3}\}.$$

Solution. Define the following values:

- Let A_n be the number of sequences of length n whose final values are of the form $\{a, b, c\}$.
- Let B_n be the number of sequences of length n whose final values are of the form $\{a, a, b\}$.
- Let C_n be the number of sequences of length n whose final values are of the form $\{a, b, a\}$.
- Let D_n be the number of sequences of length n whose final values are of the form $\{b, a, a\}$.
- Let E_n be the number of sequences of length n whose final values are of the form $\{a, a, a\}$.

In search of a recursive formula, consider the following:

- $A_{n+1} = A_n$ since all three terms must be present.
- $B_{n+1} = D_n$ since in order to get (a, a, b) , then the last three terms of the n -th sequence is (X, a, a) . We require a b to be in this set, so $X = b$ and the last three terms is (b, a, a) .
- $C_{n+1} = A_n + B_n + C_n$. We require the previous three terms to be (X, a, b) . We require there to be a a in the sequence, so that the next term is a . Clearly, this is satisfied, so $X \in \{a, b, c\}$.
- $D_{n+1} = A_n + B_n + C_n$. We require the previous three terms to be (X, b, a) . We require there to be a b in the sequence, so that the next term is b . Clearly, this is satisfied, so $X \in \{a, b, c\}$.
- $E_{n+1} = E_n + D_n$. We require the previous three terms to be (X, a, a) . We require there to be a a in the sequence, so that the next term is a . Clearly, this is satisfied, so $X \in \{a, b, c\}$. Note that there is only one D_n since the previous n -th terms are fixed.

Note that $A_3 = 1$ and $B_3 = C_3 = D_3 = E_3 = 0$. In addition,

$$B_{n+1} = D_n = A_{n-1} + B_{n-1} + C_{n-1} = C_n.$$

Therefore, substituting,

$$C_{n+1} = A_n + B_n + C_n = 1 + C_n + C_{n-1}.$$

Notice, $C_n = F_{n-1} - 1$ satisfies this recurrence. In addition, $B_n = F_{n-2} - 1$. Then,

$$E_n = E_3 + \sum_{i=3}^{n-1} D_i = \sum_{i=3}^{n-1} C_i = \sum_{i=3}^{n-1} (F_{i-1} - 1) = \sum_{i=2}^{n-2} (F_i - 1) = (F_n - 2) - (n - 3) = F_n - n + 1.$$

Then, the total sum is equivalent to:

$$\begin{aligned} \mathcal{S}_n &= A_n + B_n + C_n + D_n + E_n \\ &= 1 + (F_{n-2} - 1) + (F_{n-1} - 1) + (F_{n-1} - 1) + (F_n - n + 1) \\ &= F_n + 2F_{n-1} + F_{n-2} - n - 1. \\ &= F_{n+2} - n - 1. \end{aligned}$$

Thus, by Binet's Formula, then

$$\mathcal{S}_n = F_{n+2} - n - 1 = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n+2} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{n+2} - n - 1.$$

3. (2025 FLSAM PUMaC Tryout, Jacob Khohayting) A sequence $\{a_n\}_{n \geq 1}$ of positive integers is defined recursively by $a_1 = k$ and

$$\begin{cases} a_{n+1} = \frac{1}{2} \cdot a_n & \text{if } a_n \equiv 0 \pmod{2} \\ a_{n+1} = a_n + 77 & \text{if } a_n \equiv 1 \pmod{2} \end{cases}$$

for all n . Determine, with proof, all values of k for which $a_6 = a_1$.

Solution. Note that $a_1 \equiv a_6 \pmod{77}$. Note the second operation does not change $a_i \pmod{77}$. Now, we will compute the number of times the first operation is applied. To do this, consider the sequence of remainders modulo 7 and 11 obtained by only applying the first condition.

$$A \longrightarrow 4A \longrightarrow 2A \longrightarrow A \pmod{7}$$

$$A \longrightarrow 6A \longrightarrow 3A \longrightarrow 7A \longrightarrow 9A \longrightarrow 10A \longrightarrow 5A \longrightarrow 8A \longrightarrow 4A \longrightarrow 2A \longrightarrow A \pmod{11}.$$

Thus, we require $a_1 \equiv 0 \pmod{11}$. In addition, either $a_1 \equiv 0 \pmod{7}$ or we apply the first operation three times and the second operation twice.

If $a_1 \equiv 0 \pmod{77}$, one can scale down by 77 to remove this condition. So, define $b_i = \frac{a_i}{77}$. Then, $b_{n+1} = \frac{1}{2} \cdot b_n$ if $b_n \equiv 0 \pmod{2}$ and $b_{n+1} = b_n + 1$ if $b_n \equiv 1 \pmod{2}$. Notice, if we apply the second operation (2), we must apply the first operation (1) immediately. Consider casework on parity of the first number.

- If $b_1 \equiv 0 \pmod{2}$, then we must apply the first operation (1). From here, we have a few cases to consider:

$$(1)(1)(2)(1)(2) \implies X \rightarrow \frac{X}{2} \rightarrow \frac{X}{4} \rightarrow \frac{X}{4} + 1 \rightarrow \frac{X}{8} + \frac{1}{2} \rightarrow \frac{X}{8} + \frac{3}{2} \implies X = \frac{12}{7}$$

$$(1)(2)(1)(1)(2) \implies X \rightarrow \frac{X}{2} \rightarrow \frac{X}{2} + 1 \rightarrow \frac{X}{4} + \frac{1}{2} \rightarrow \frac{X}{8} + \frac{1}{4} \rightarrow \frac{X}{8} + \frac{5}{4} \implies X = \frac{10}{7}.$$

$$(1)(2)(1)(2)(1) \implies X \rightarrow \frac{X}{2} \rightarrow \frac{X}{2} + 1 \rightarrow \frac{X}{4} + \frac{1}{2} \rightarrow \frac{X}{4} + \frac{3}{2} \rightarrow \frac{X}{8} + \frac{3}{4} \implies X = \frac{6}{7}.$$

Notice, (2)(1)(1)(1)(2) is impossible as by applying the (1)(2) at the end, then $b_6 \equiv 0 \pmod{2}$, a contradiction. None of these cases are satisfactory as X is not an integer in any case.

- If $b_1 \equiv 1 \pmod{2}$, then we must apply the first operation (2). From here, we have a few cases to consider:

$$(2)(1)(2)(1)(1) \implies X \rightarrow X + 1 \rightarrow \frac{X}{2} + \frac{1}{2} \rightarrow \frac{X}{2} + \frac{3}{2} \rightarrow \frac{X}{4} + \frac{3}{4} \rightarrow \frac{X}{8} + \frac{3}{8} \implies X = \frac{3}{7}.$$

$$(2)(1)(1)(2)(1) \implies X \rightarrow X + 1 \rightarrow \frac{X}{2} + \frac{1}{2} \rightarrow \frac{X}{4} + \frac{1}{4} \rightarrow \frac{X}{4} + \frac{5}{4} \rightarrow \frac{X}{8} + \frac{5}{8} \implies X = \frac{5}{7}.$$

None of these cases are satisfactory as X is not an integer in any case.

Suppose we were to apply the first operation three times and the second operation twice. Note that these cases are also just scaling the cases from before back up by 77 as they have three (1)'s and two (2)'s, but we must check to ensure X is divisible by 11 and satisfies all other conditions.

- If $a_1 \equiv 0 \pmod{2}$, then we require the following cases:

$$(1)(1)(2)(1)(2) : 132 \rightarrow 66 \rightarrow 33 \rightarrow 110 \rightarrow 55 \rightarrow 132$$

$$(1)(2)(1)(1)(2) : 110 \rightarrow 55 \rightarrow 132 \rightarrow 66 \rightarrow 33 \rightarrow 110$$

$$(1)(2)(1)(2)(1) : 66 \rightarrow 33 \rightarrow 110 \rightarrow 55 \rightarrow 132 \rightarrow 66$$

It is easy to confirm all these work.

- If $a_1 \equiv 1 \pmod{2}$, then we require the following cases:

$$(2)(1)(2)(1)(1) : 33 \rightarrow 110 \rightarrow 55 \rightarrow 132 \rightarrow 66 \rightarrow 33$$

$$(2)(1)(1)(2)(1) : 55 \rightarrow 132 \rightarrow 66 \rightarrow 33 \rightarrow 110 \rightarrow 55$$

Notice, $(2)(1)(1)(1)(2)$ is impossible as by applying the $(1)(2)$ at the end, then $a_6 \equiv 0 \pmod{2}$, a contradiction. It is easy to confirm all these work.

Thus, the sum is $132 + 110 + 66 + 33 + 55 = \boxed{396}$.